

Preprocessing & Feature Engineering Report

Economic Forecasting for Banking Using FRED Time Series: Gradient Boosting with Phased Risk Scenario Analysis

Author: Patrick Lynch

Team name: Dragon Exploder

GitHub repository: https://github.com/lynchpo/dragon_exploder_fred

Recap: Background & Question

Research question: Can a carefully engineered set of macroeconomic time series from the Federal Reserve Economic Data (FRED) repository be used to produce accurate multi-step forecasts of the 30-year fixed mortgage rate while also generating scenario-based risk metrics that support decision-making under uncertainty?

Hypothesis: A carefully selected combination of leading macroeconomic indicators contains predictive information about future movements in the 30-year fixed mortgage rate because these indicators capture the primary channels through which monetary policy and real-economy conditions influence long-term borrowing costs.

Prediction: A gradient-boosting model that incorporates engineered lag, rolling, and yield-curve features will achieve lower out-of-sample forecasting error than a SARIMAX baseline. In a later phase, Monte Carlo paths will support calculation of risk metrics such as maximum drawdown.

Methods

In the Exploratory Data Analysis report, a concrete preprocessing agenda was outlined: align all series to a common weekly frequency, forward-fill monthly indicators after resampling, respect publication lags, construct lag and rolling features, create a yield-curve spread, and perform any feature screening inside the training folds of time-series cross-validation. The present work implements that agenda and reports the results.

Preprocessing Methods and Justifications

All six core series were restricted to the period beginning January 2000 and resampled to a weekly frequency ending on Fridays. The mortgage-rate series is already weekly; the daily 10-Year Treasury series was aggregated by taking the last available value in each week. Monthly series (Federal Funds rate, unemployment, CPI, and Housing

Starts) were first resampled to weekly frequency and then forward-filled. Forward-fill was chosen because it carries the last known official value until a new observation arrives, which mirrors the information set that would have been available to a forecaster in real time. Linear interpolation was considered and rejected because it would invent intermediate values that did not exist at the time. After alignment, a small number of missing values remained at the very beginning of a few monthly series (eight observations). These were eliminated when lag and rolling features were subsequently created and rows containing NaNs were dropped.

A time-ordered train/test split was used rather than a random split. The first 80 percent of the aligned sample (24 March 2000 through 5 March 2021, 1,094 weekly observations) forms the training set; the final 20 percent (12 March 2021 through 5 June 2026, 274 observations) forms the test set. This chronological split respects the temporal structure of the data and prevents leakage of future information into model training. The mean of the target variable is 5.04 percent in the training window and 5.83 percent in the test window, reflecting the higher-rate environment of the post-2021 period. After the split, both partitions contain zero missing values.

Feature Engineering Methods and Justifications

Three families of engineered features were created. First, a yield-curve spread was defined as the difference between the 10-Year Treasury rate and the Federal Funds rate. This spread is a classic indicator of the term premium and of monetary-policy stance. Second, lagged values (1-week, 2-week, and 4-week) were constructed for the mortgage rate itself, the 10-Year Treasury, the Federal Funds rate, and the yield-curve spread. These lags supply the short-term momentum information that tree-based models require. Third, rolling means and standard deviations (4-week and 12-week windows) were computed for the mortgage rate, the 10-Year Treasury, and the Federal Funds rate. Rolling statistics capture local trend and local volatility. After these transformations, any row that still contained a missing value (arising from the lag and rolling calculations) was dropped, leaving 1,368 complete weekly observations and 27 predictor features plus the target.

All lag and rolling features were generated on the full aligned series before the train/test cut. In the subsequent modeling phase, any further feature screening or importance ranking will be performed strictly inside each training fold of a time-series cross-validation loop so that no information from the test period or from future folds can influence feature selection.

Results

Cleaning, Alignment, and Split Outcomes

After frequency alignment and forward-fill, the working data set contained 1,387 weekly rows and six original series. Construction of lag and rolling features reduced the sample to 1,368 complete rows. The chronological 80/20 split produced a training set of 1,094 observations and a test set of 274 observations. Both partitions are free of missing values. These results confirm that the preprocessing pipeline described in the EDA report is operational and yields a clean, analysis-ready matrix.

Unsupervised Feature Exploration (PCA)

Principal Component Analysis was applied to the 27 engineered predictors (after standardization) on the training set. The cumulative explained-variance curve rises steeply: approximately three to four principal components capture 90 percent of the total variance, and fewer than ten components capture nearly all of it. This pattern indicates that the engineered feature set is highly redundant—an expected outcome when multiple lags and rolling statistics are derived from the same underlying series. Because the original predictors already carry clear economic meaning (Treasury yields, policy rate, unemployment, inflation, housing activity), collapsing them into abstract principal components would reduce interpretability for risk and treasury stakeholders without a commensurate gain in forecasting signal. Consequently, PCA is retained only as a diagnostic and is not used as a dimensionality-reduction step in the primary modeling pipeline. The high redundancy does, however, reinforce the need for careful regularization or importance-based pruning when gradient-boosting models are trained.

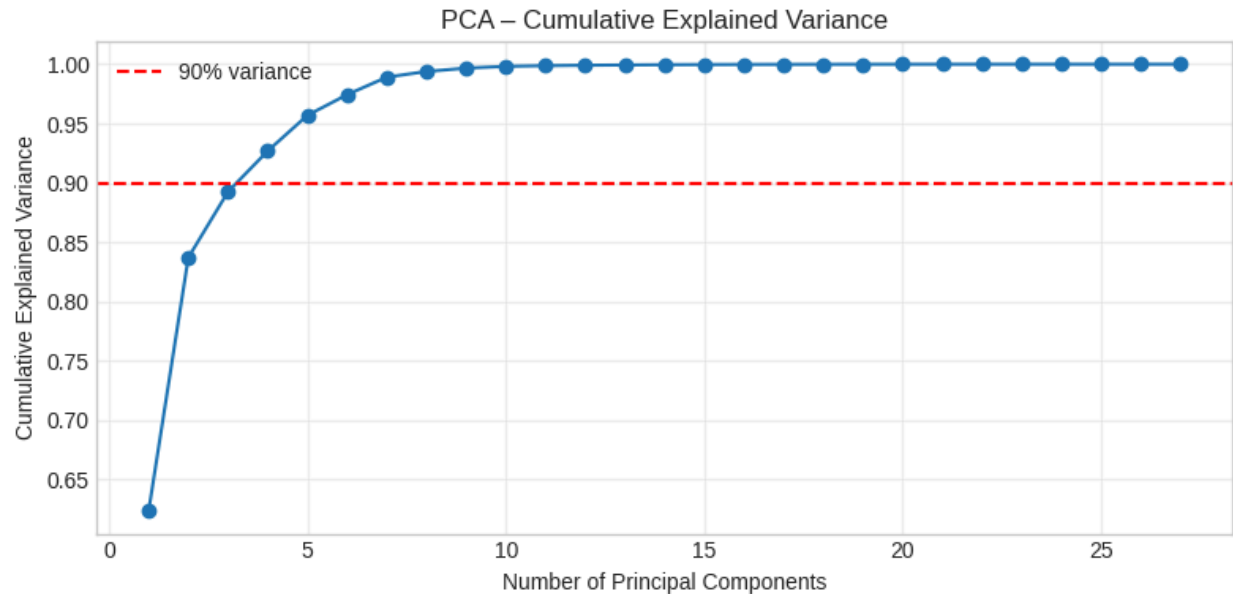


Figure 1. Cumulative explained variance from PCA on the 27 engineered predictors (training set). Roughly three to four components reach the 90 percent threshold, confirming substantial redundancy among lag and rolling features.

Supervised Feature Selection Plan

Supervised feature selection will be performed after the first gradient-boosting models are trained. Two complementary approaches are planned. First, built-in feature-importance measures (gain or split count) from XGBoost or LightGBM will be examined on each training fold of a time-series cross-validation scheme. Second, a simple recursive feature-elimination loop may be applied inside the same folds if the initial importance rankings suggest that a substantial fraction of the 27 features contribute little predictive power. All selection decisions will be locked before the final test-set evaluation so that the test period remains untouched. This plan keeps the economic interpretability of the original and engineered series while allowing the data to prune redundant lags and rolling statistics.

A correlation matrix of selected engineered features (target, key levels, lags, and rolling means) confirms the expected pattern: lags and rolling means of the same series are almost perfectly correlated with one another and with the contemporaneous level. The yield-curve spread shows a strong negative association with the Federal Funds rate, as expected by construction. These diagnostics further support the decision to rely on supervised importance ranking rather than unsupervised compression.

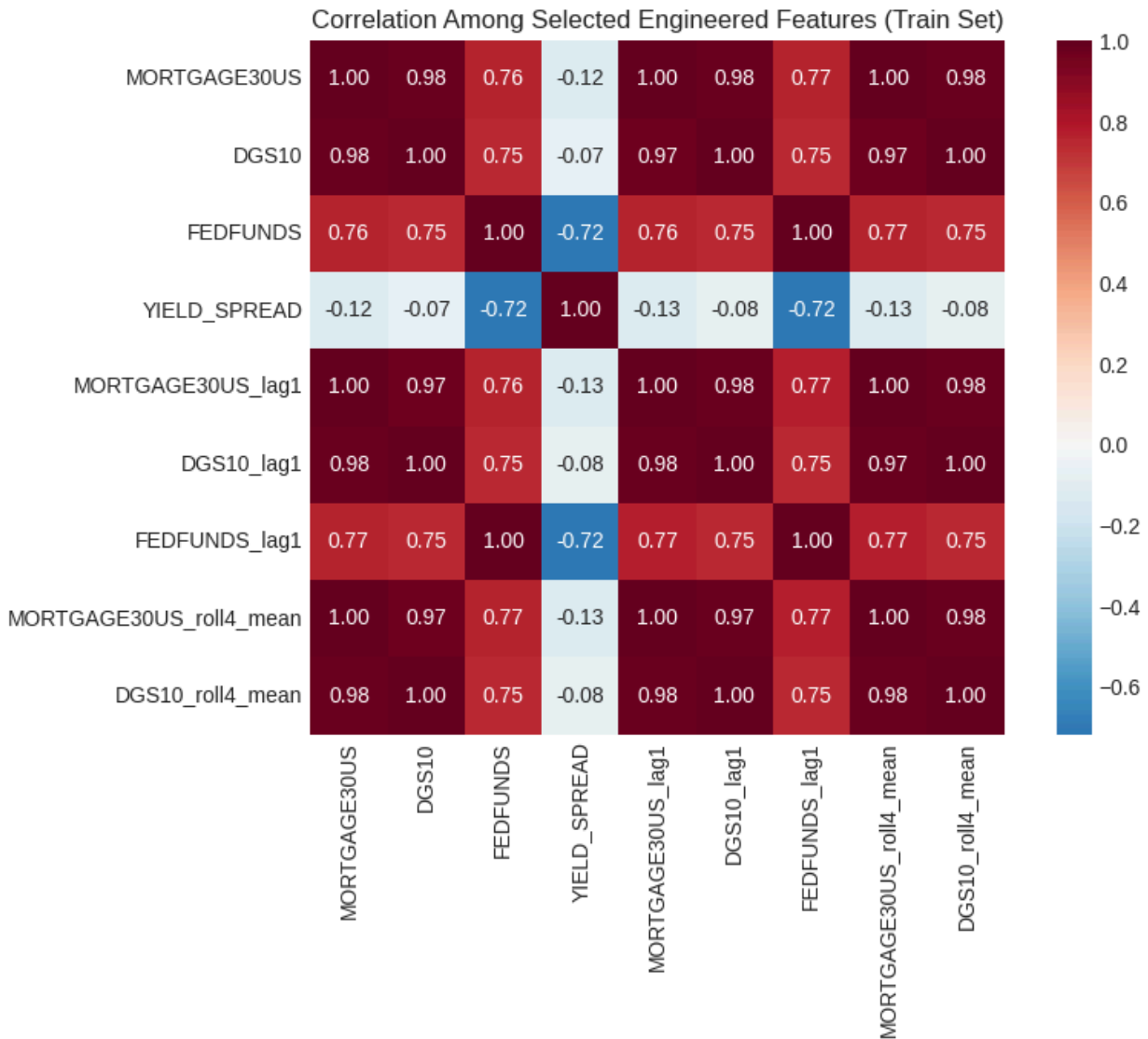


Figure 2. Correlation among a subset of engineered features on the training set. High collinearity among lags and rolling means of the same series is evident and will be addressed through supervised importance ranking.

Discussion & Next Steps

Key Takeaways

The preprocessing pipeline has converted six mixed-frequency FRED series into a clean weekly matrix of 1,368 observations and 27 predictor features. Frequency alignment with forward-fill, chronological train/test splitting, and the construction of lag, rolling, and spread features were all completed successfully and left no missing values in either partition. Principal Component Analysis revealed high redundancy among the engineered features, confirming that unsupervised dimensionality reduction would

sacrifice interpretability for little gain; therefore PCA is used only diagnostically. The original research question and hypothesis remain unchanged: the data continue to support the expectation that a gradient-boosting model using these engineered features can improve upon a classical baseline. The only material revision to the earlier analysis plan is the explicit decision to de-emphasize PCA as a modeling step and to rely instead on supervised feature-importance ranking performed inside time-series cross-validation folds.

Modeling Plan

The next phase of work will proceed as follows:

1. Baseline model: Fit a SARIMAX model on the training set using the aligned weekly mortgage-rate series and evaluate multi-step forecasts on the test set with RMSE and MAE.
2. Gradient-boosting models: Train XGBoost and/or LightGBM regressors on the full set of 27 engineered features, using time-series cross-validation (e.g., expanding or sliding windows) for hyper-parameter tuning and for supervised feature-importance ranking.
3. Feature pruning: Retain only those features that consistently rank high across cross-validation folds; re-train the preferred model on the reduced feature set.
4. Final evaluation: Compare the pruned gradient-boosting model against the SARIMAX baseline on the held-out test period using RMSE, MAE, and directional accuracy.
5. Interpretation: Use SHAP values on the final model to confirm that the economically motivated predictors (Treasury yields, policy rate, lags, and spreads) contribute meaningfully to the forecasts.

This sequence keeps the project inside the Phase-1 scope agreed with the instructor while producing a transparent, reproducible comparison between classical and modern forecasting approaches.

Appendix: Data Dictionary (Updated)

The dictionary below reflects the series and engineered features present after preprocessing. The target remains MORTGAGE30US.

Variable	Description	Type	Role
MORTGAGE30US	30-Year Fixed Mortgage Rate (aligned weekly)	float	Target
DGS10	10-Year Treasury Rate (weekly last)	float	Predictor
FEDFUNDS	Federal Funds Rate (weekly, forward-filled)	float	Predictor
UNRATE	Unemployment Rate (weekly, forward-filled)	float	Predictor
CPIAUCSL	Consumer Price Index (weekly, forward-filled)	float	Predictor

HOUST	Housing Starts (weekly, forward-filled)	float	Predictor
YIELD_SPREAD	DGS10 – FEDFUNDS	float	Engineered
*_lag1 / lag2 / lag4	1-, 2-, and 4-week lags of selected series	float	Engineered
*_roll4_mean / std	4-week rolling mean and standard deviation	float	Engineered
*_roll12_mean	12-week rolling mean	float	Engineered

Summary Statistics of the Processed Data

Aligned weekly rows (before feature engineering): 1,387

Complete rows after lag/rolling features: 1,368

Training observations: 1,094 (2000-03-24 to 2021-03-05)

Test observations: 274 (2021-03-12 to 2026-06-05)

Number of predictor features: 27

Missing values in train and test after processing: 0